2.5 Nanomaterials

Problems Worksheet

1. Convert the following lengths to nanometres:

    a. 33.5 mm

    b. 960 cm

    c. 0.42 m

    d. 250 µm

2. Define nanomaterials.

3. Give two examples of naturally occurring nanomaterials.

4. Nanomaterials can be produced in two different ways. Describe these.

5. Composite materials are often used to manufacture goods.

    a. Describe what is meant by a composite material.

    b. What is a nanocomposite?

6. Nanomaterials often have different properties to the same type of matter made up of larger particles.

    a. Why do nanoparticles often have a different appearance to the bulk material?

    b. Explain why nanoparticles can be useful as catalysts.

7. Nanoparticles can be useful as the active ingredient in sunscreens.

    a. Explain why nanoparticle sunscreen is transparent even though it still contains zinc oxide and titanium dioxide, both of which are white and opaque.

    b. How is nanoparticle sunscreen effective?

8. Explain the possible implications of nanoparticles on health and the environment.